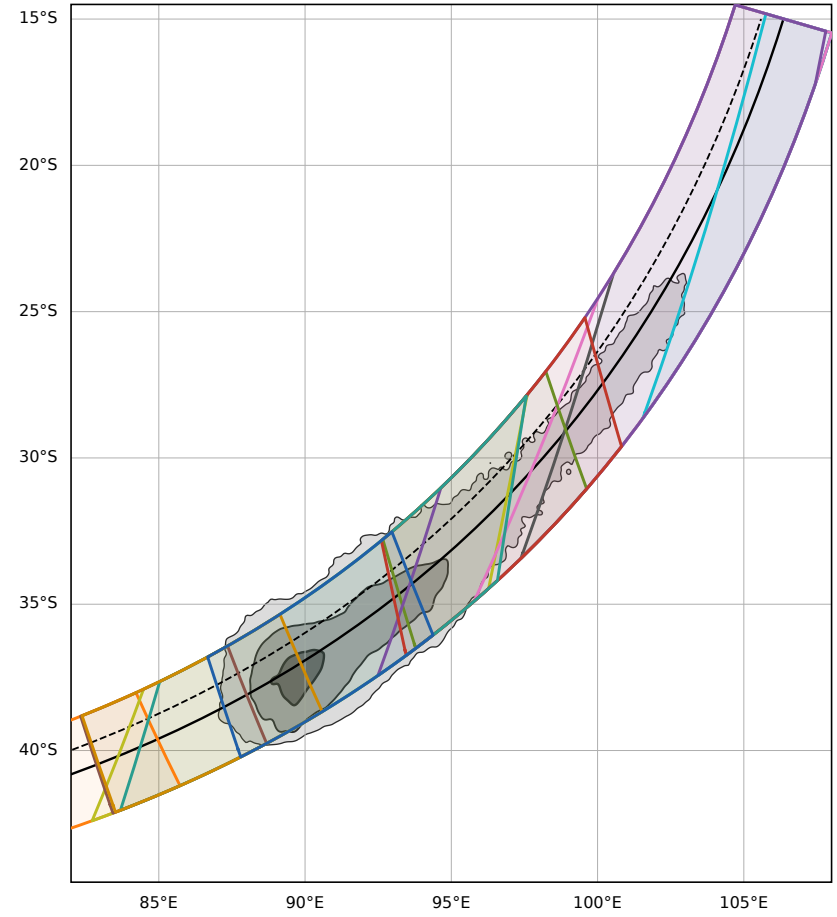
Satellite passes that could have imaged the impact area near the time of impact



- COSMO-SkyMed 3, 00:35-00:36 UTC, radar
- COSMO-SkyMed 2, 00:54-00:55 UTC, radar
- RADARSAT-2, 23:57-00:02 UTC, radar
- Beijing-1, 00:52-00:55 UTC, optical, daylight
- Yaogan-11, 01:11-01:18 UTC, optical, daylight
- COSMO-SkyMed 1, 00:07-00:09 UTC, radar
- Yaogan-8, 01:35-01:41 UTC, optical, daylight
- KOMPSAT-5, 01:01-01:02 UTC, radar
- KH-11 (USA 186), 00:54-00:58 UTC, optical, daylight
- Yaogan-4, 02:07-02:11 UTC, optical, daylight
- Cartosat-2, 02:02-02:08 UTC, optical, daylight
- COSMO-SkyMed 4, 01:18-01:19 UTC, radar
- 50 % of impact probability
- 90 % of impact probability
- 99 % of impact probability
- 7th arc (00:19, sea level)
- 6th arc (00:11, sea level)
- 100 NM either side of the 7th arc

Share of impact probability inside the band

Satellite	30 min before	0-60 min after
COSMO-SkyMed 3	0.347	0.582
COSMO-SkyMed 2	<0.001	0.494
RADARSAT-2	0.427	0.000
Beijing-1	<0.001	0.092
Yaogan-11	0.000	0.085
COSMO-SkyMed 1	0.055	0.000
Yaogan-8	0.000	0.006
KOMPSAT-5	<0.001	0.005
KH-11 (USA 186)	<0.001	0.001
Yaogan-4	0.000	<0.001
Cartosat-2	0.000	<0.001
COSMO-SkyMed 4	0.000	<0.001

Any radar or daylight optical, 0-60 min after: 0.855

(i) Coloured areas show where each satellite could point its sensor, cut to 100 NM either side of the 7th arc. Grey shows where the aircraft probably hit the sea. An area is cleared only if (a) the image exists, (b) the target can be detected in it, and (c) somebody examined it. This map does not show that any image was taken.

(ii) Impact density: end-of-flight run J, waypoints family (core P = 0.9982), 24 seeds, 1.5 M draws from the rebuilt flight-posterior weights, 00:19 option '00:19 data not used' (D0 total, q = 1/3, other log-on cause silent); 0.05 deg histogram, Gaussian sigma 0.1 deg, highest-density regions. Arcs: shared sea-level arcs. Footprints: SGP4 on March 2014 element sets; SAR band = near-far incidence on the allowed look side(s); optical band = max off-nadir + half swath, sun >= 10 deg at the point. Passes kept if the band covered any impact draw from 30 min before to 60 min after that draw's impact. 'Before' crossings test a fixed point, not the moving aircraft. One SAR scene covers only one beam (about 1-3 % of the band in nominal modes). 19 classified satellites without element sets are not included. KH-11 (USA 186) element set is 210 h from the pass, so its timing is uncertain by minutes.